

Diabetes, Hypertension and Stroke Prediction Using Supervised Learning Model

Abstract

Diabetes, strokes, and hypertension are all serious health conditions that can have significant impacts on an individual's overall health and well-being. Diabetes is a chronic condition characterized by high levels of sugar (glucose) in the blood, which can lead to a range of complications including heart disease, kidney disease, nerve damage, and eye problems. Strokes are caused by an interruption of blood flow to the brain, which can lead to brain damage, paralysis, and other serious complications. Hypertension, or high blood pressure, is a condition in which the blood flows through the arteries at a higher than normal pressure, which can increase the risk of heart attack, stroke, and other health problems. Machine learning techniques have been applied to the study of these diseases to predict the risk of diabetes, stroke, and hypertension in individuals based on various risk factors such as age, gender, BMI, and family history. In this project, we have used a supervised learning model of ML. From the supervised model, we have specifically used Logistic regression, Gaussian Naive Bayes, Bernoulli Naive Bayes, KNN and Decision tree algorithms. With all these methods, we are trying to train our model in the best way possible to predict the possibility of diabetes, stroke and hypertension of a test sample with higher accuracy. Overall, the use of this project in the study of diabetes, strokes, and hypertension has the potential to improve diagnosis, treatment, and management of these conditions, leading to improved patient outcomes.

Introduction

Stroke, diabetes, and hypertension are three important health conditions that can have serious consequences if not properly managed. Classification using machine learning can be a useful tool for predicting and identifying individuals at risk for these conditions, as well as for monitoring and managing the progress of treatment.

In the context of machine learning, classification refers to the process of training a model to assign input data to one of several predefined categories, or classes. This can be done using a variety of algorithms, including decision trees, support vector machines, and neural networks.

For example, a machine learning model could be trained to classify individuals as either having or not having a stroke based on a range of input features, such as age, blood pressure, and cholesterol levels. Similarly, a model could be trained to predict the likelihood of an individual developing diabetes or hypertension based on factors like weight, family history, and lifestyle habits.

One potential benefit of using machine learning for classification of stroke, diabetes, and hypertension is that it can enable more accurate and efficient identification of individuals at risk, potentially leading to earlier intervention and better outcomes. However, it is important to note

that machine learning is only one tool among many that can be used for these purposes, and the accuracy of any particular model will depend on the quality and relevance of the data used to train it.

Classification of stroke, diabetes, and hypertension using machine learning is important because these conditions are major public health issues that can have serious consequences if not properly managed. Stroke, diabetes, and hypertension are all chronic conditions that can lead to other health problems, including heart disease, kidney disease, and nerve damage. By using machine learning to classify these conditions, it is possible to identify individuals who may be at risk for these conditions and provide them with the appropriate care and treatment. This can help prevent or mitigate the negative effects of these conditions, improving the overall health and well-being of the population.

In addition, machine learning can be used to identify patterns and trends in the data related to these conditions, which can help to inform public health policy and strategies for preventing and managing these conditions. This can lead to more effective interventions and better outcomes for individuals with these conditions.

Overall, the use of machine learning in the classification of stroke, diabetes, and hypertension can help to improve the diagnosis and management of these conditions, leading to better health outcomes for individuals and populations.

Previous work

There has been significant research on using machine learning to classify stroke, diabetes, and hypertension. Here are a few examples:

Stroke classification:

A study published in the Journal of Neuroimaging in 2020 used machine learning to classify stroke subtypes (ischemic versus hemorrhagic) based on computed tomography (CT) scans. The researchers used a convolutional neural network (CNN) and achieved an accuracy of 89.6%.

Diabetes classification:

A study published in the Journal of Diabetes Research in 2018 used machine learning to predict the onset of type 2 diabetes in a high-risk population. The researchers used a random forest classifier and achieved an accuracy of 77.5%.

Hypertension classification:

A study published in the Journal of Clinical Hypertension in 2018 used machine learning to predict hypertension in a population of African Americans. The researchers used a support vector machine (SVM) and achieved an accuracy of 81.6%.

It's important to note that these are just a few examples, and the performance of machine learning models for these tasks can vary depending on the specific data and methods used.

A study published in the Journal of Medical Systems in 2018 that used machine learning algorithms to classify stroke patients based on their clinical characteristics and imaging data. The study found that the machine learning model had high accuracy and could be used as a tool for predicting stroke outcomes.

A study published in the Journal of Diabetes Science and Technology in 2015 that used machine learning algorithms to predict the risk of developing diabetes in a population. The study found that the machine learning model was able to accurately predict the risk of developing diabetes and could be used as a tool for early diagnosis and prevention.

A study published in the Journal of Clinical Hypertension in 2016 that used machine learning algorithms to predict the likelihood of developing hypertension in a population. The study found that the machine learning model was able to accurately predict the risk of developing hypertension and could be used as a tool for early diagnosis and prevention.

Methodology

Workflow diagram of the project

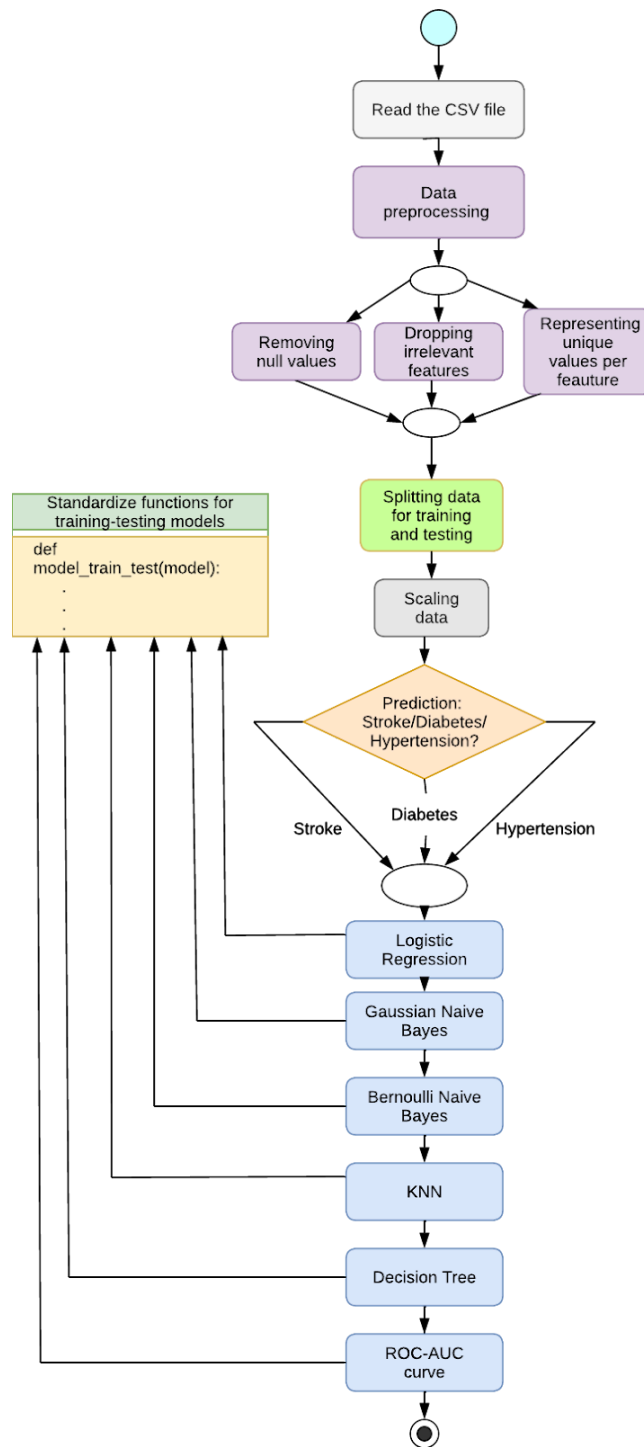


Figure 1: Workflow diagram

Dataset

The dataset of our project consists of 70,692 data. There are 18 feature variables and 3 target variables, Diabetes, Hypertension, and Stroke, each target variable having 2 classes. This dataset is balanced. Excluding the target variables, the feature variables include age, sex, high cholesterol, cholesterol check, BMI, smoker, heart disease, physical activity, fruits, veggies, heavy alcohol consumption, general health, diffwalk.

Preprocessing Steps

- Null values have been deducted.
- Unique values have been stored in the data frame.
- Unnecessary features have been dropped after analyzing the data set.
- Finally, the data is normalized with a standard scaler.

Data Visualization

Initially we have represented categorical data through bar charts. Then to display trends in data, heatmap is used. Here a warm-to-cool schema for their visual representation. As a result, warmer colors (violate) show higher activity while cooler colors (orange) represent lower activity

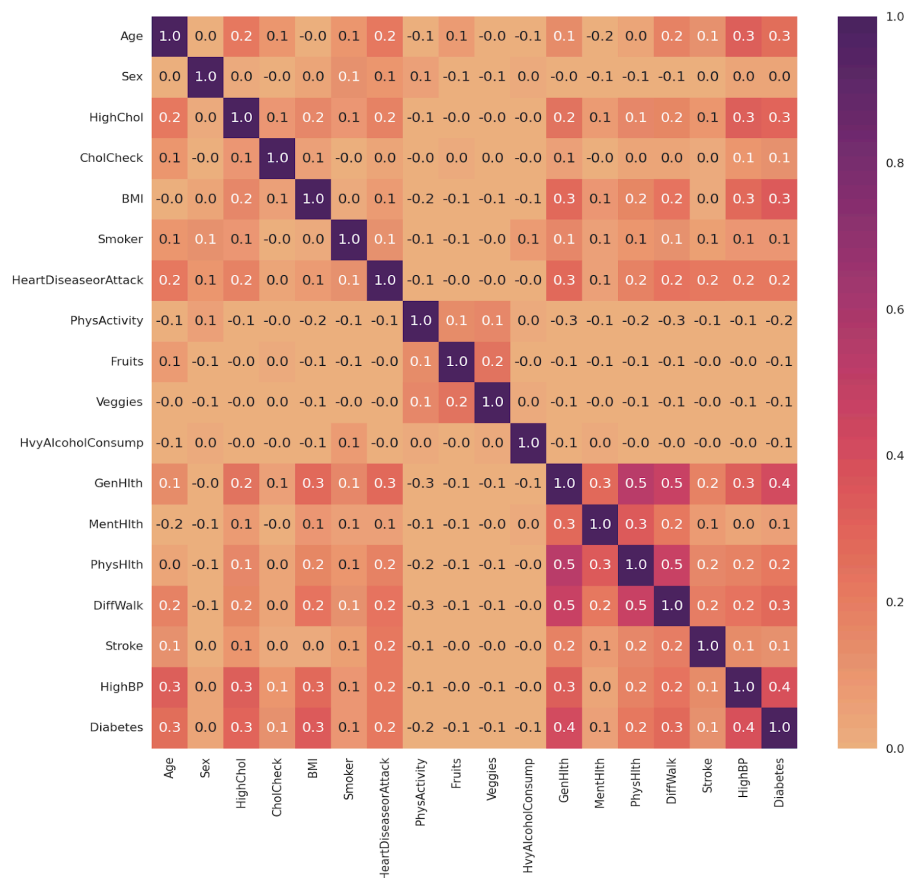


Figure 3: Data representation using Heatmap

Model Implementation and Algorithms

In this project basically Supervised Machine Learning is used in which we have used labeled datasets for training the model, making predictions of the output values and comparing its output with other outputs, and then compute the errors to modify the model accordingly. Also, as the system is trained enough using this learning method, it becomes capable enough to provide the target values from any new input.

From the supervised model we are specifically using classification where the output variable is discrete. i.e. we train the hypothesis($f(x)$) in a way to get discrete output(y) for the input data(x). The output can also be termed as a class.

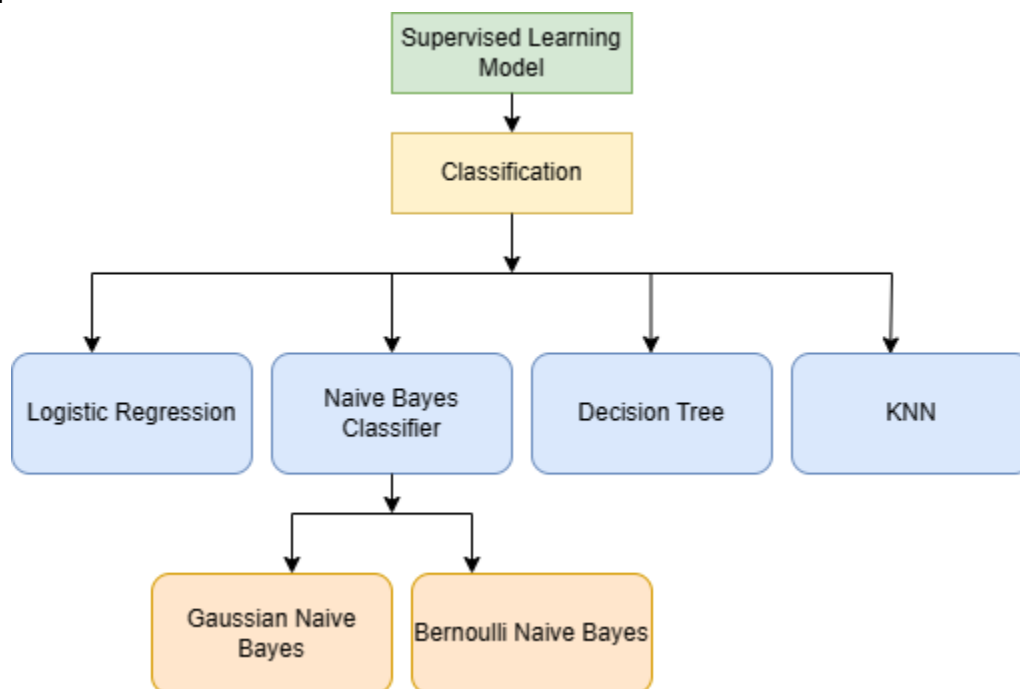


Figure 2 : Model and algorithms

All the algorithms that we have used are mainly classification algorithms. These are: Logistic regression, Naive Bayes classifier (Gaussian, Bernoulli), Decision tree and KNN.

Classification is a machine learning task that involves predicting a categorical label for a given data point. In the context of stroke, diabetes, and hypertension, classification algorithms could be used to predict the likelihood that an individual has one of these conditions based on certain features such as age, blood pressure, cholesterol levels, and family history.

There are several different classification algorithms that could be used to predict stroke, diabetes, and hypertension, including:

Logistic Regression: This is a type of regression algorithm that is used for classification tasks. It works by fitting a logistic curve to the data and using this curve to predict the probability that a given data point belongs to a particular class.

Gaussian Naive Bayes: This is a probabilistic classifier that is based on the assumption that the features of a data point are independent of one another and follow a Gaussian distribution. It uses Bayes' Theorem to calculate the probability that a given data point belongs to a particular class.

Bernoulli Naive Bayes: This is similar to Gaussian Naive Bayes, but it is used for data that is binary (i.e., it consists of only two possible values). It works by calculating the probability that a given data point belongs to a particular class based on the presence or absence of certain features.

K-Nearest Neighbors (KNN): This is a non-parametric classification algorithm that works by finding the k data points that are most similar to a given data point and using the majority class among those k points to classify the data point.

Decision Tree: This is a tree-based classification algorithm that works by recursively partitioning the data into smaller and smaller groups based on a set of decision rules. It creates a tree-like model of decisions and uses it to make predictions.

These are just a few of the many classification algorithms that could be used to predict stroke, diabetes, and hypertension. Each algorithm has its own strengths and weaknesses, and the choice of which algorithm to use will depend on the specific characteristics of the data and the goals of the analysis.

Result Analysis

In this report to represent the result we have used confusion matrices, ROC curves. In addition we have mentioned precision, accuracy, recall and f1- score for each model in a table for each prediction segment (Stroke, Hypertension and Diabetes). Confusion matrix is a table that summarizes how successful the classification model is at predicting examples belonging to various classes. In definition it is defined as the actual correct prediction divided by total prediction made by model. In a classification problem with two classes, recall is calculated as the number of true positives divided by the total number of true positives and false negatives. Almost always, in practice, we have to choose between a high precision or a high recall. It's usually impossible to have both.

ROC curves commonly use the combination of true positive rate (TPR) and false positive rate (FPR). The higher the area under the ROC curve (AUC), the better the classifier. A classifier with an AUC higher than 0.5 is better than a random classifier. If AUC is lower than 0.5, then something is wrong with your model. A perfect classifier would have an AUC of 1.

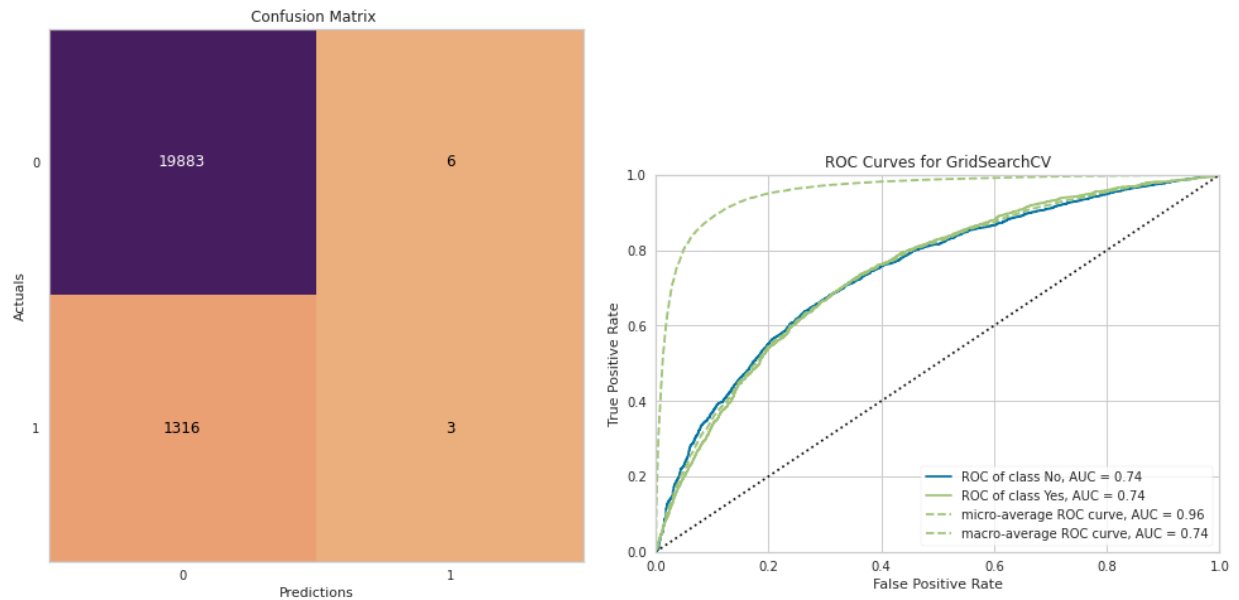
AUC values	Test quality
0.9 - 1.0	Excellent
0.8 - 0.9	Very good
0.7 - 0.8	Good
0.6 - 0.7	Satisfactory
0.5 - 0.6	Unsatisfactory

Stroke:

1) Logistic Regression

	Precision	Recall	f1-score	support
0.0	0.95	0.93	0.94	19889
1.0	0.22	0.31	0.26	1319
accuracy			0.89	21208
macro avg	0.59	0.62	0.60	21208
weighted avg	0.91	0.89	0.90	21208

In the model summary, we got a good f1-score in accuracy. It proves that logistic regression has good accuracy in detecting stroke.

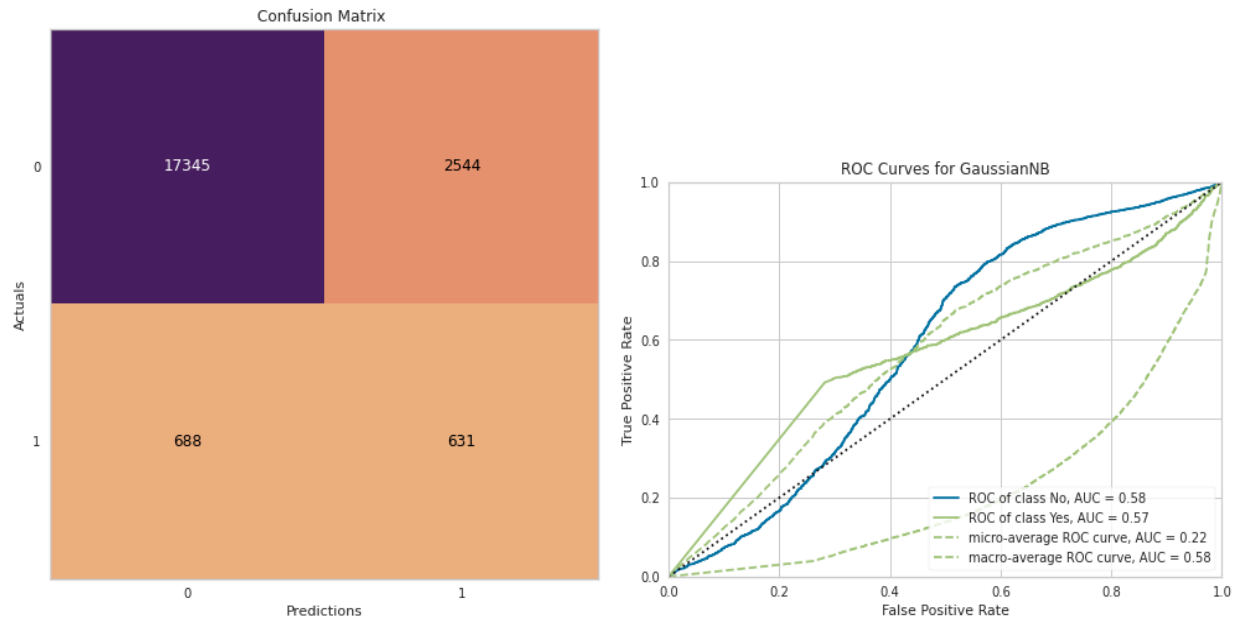


In the confusion matrix we can see among 21208 test data. We got 19883 data as true positive, 6 data as False negative, 1316 data as false positive and 3 data as true negative. In ROC curves, we get a good, very good and excellent result if we consider the above table.

2) Gaussian Naive Bayes

	Precision	Recall	f1-score	support
0.0	0.96	0.87	0.91	19889
1.0	0.20	0.48	0.28	1319
accuracy			0.85	21208
macro avg	0.58	0.68	0.60	21208
weighted avg	0.91	0.85	0.88	21208

In the model summary, we didn't get good f1-score as logistic regression. As our data is too much simplified so got good accuracy in detecting stroke but still we managed to achieve 85% f1-score in accuracy.

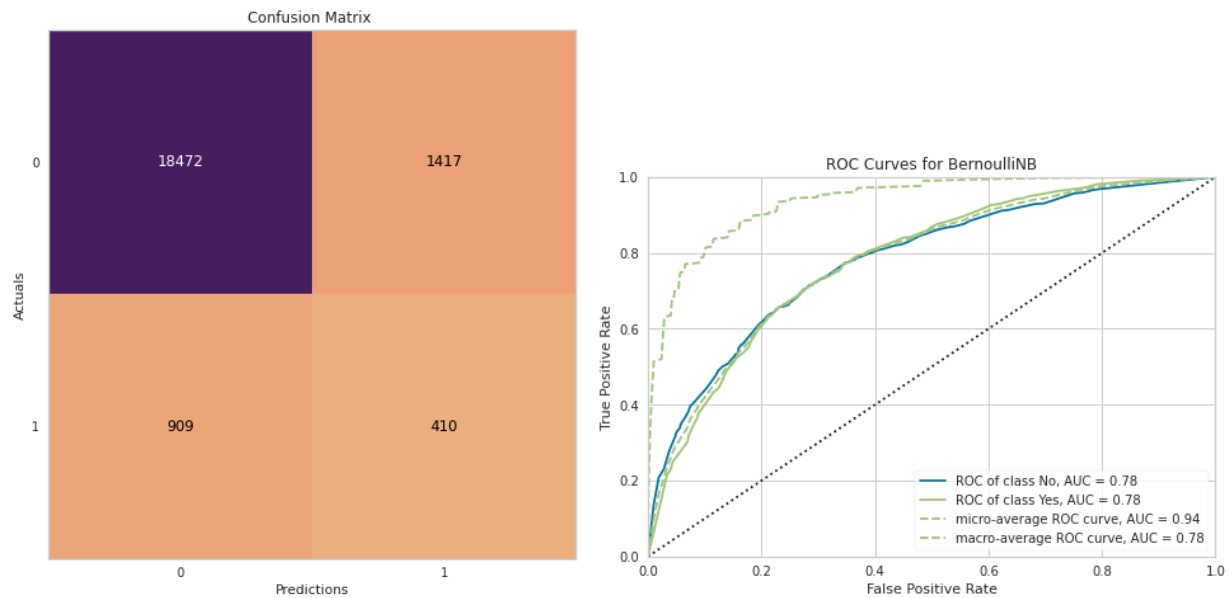


In gaussian naive bayes we got 17345 true positive, 2544 false negative, 688 false positive and 631 true negative among 21208 test data. We achieved 84.76% accuracy in detecting stroke. If we consider ROC curves, it's clear that we didn't get good result.

3) Bernoulli Naive Bayes

	Precision	Recall	f1-score	support
0.0	0.95	0.93	0.94	19889
1.0	0.22	0.31	0.26	1319
accuracy			0.89	21208
macro avg	0.59	0.62	0.60	21208
weighted avg	0.91	0.89	0.90	21208

In the model summary, we can see that we get a good f1-score - 89% in case of accuracy which is better than gaussian naive bayes.

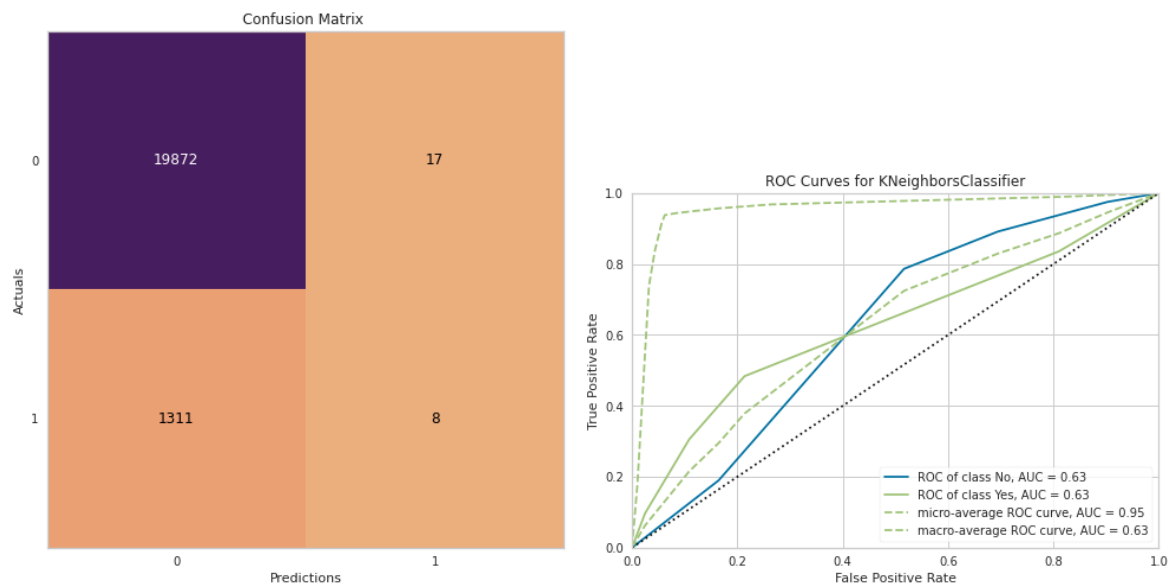


In confsuin matrix we have 18472 true positives, 1417 false negative, 909 false positive and 410 true negative among 21208 data. We got 89.03 accuracy. If we analyze the ROC curves, we have achieved a good score.

4) K-Nearest Neighbors

	Precision	Recall	f1-score	support
0.0	0.94	1.00	0.97	19889
1.0	0.32	0.01	0.01	1319
accuracy			0.94	21208
macro avg	0.63	0.50	0.49	21208
weighted avg	0.90	0.94	0.91	21208

In the model summary, we can see that we get a good f1-score - 94% in case of accuracy which is better than other models.

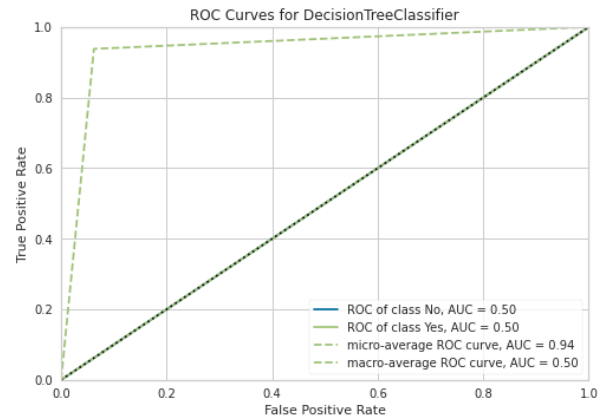
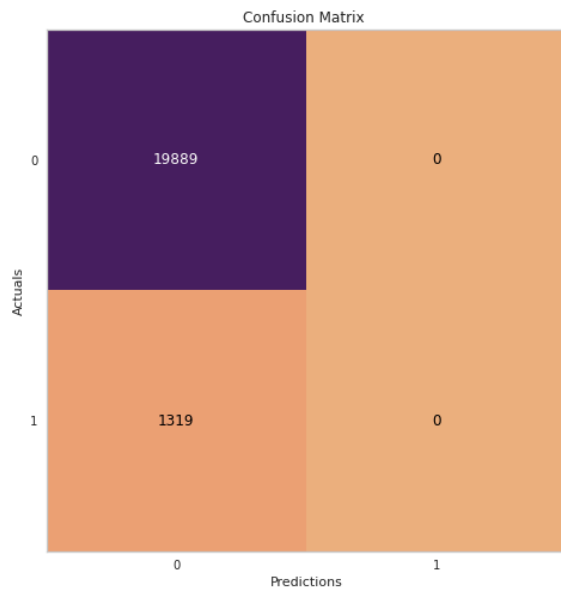


In confusion matrix we have 19872 true positives, 17 false negative, 1311 false positive and 410 true negative among 21208 data. We got 93.74 accuracy. If we analyze the ROC curves, we have achieved a good score.

5) Decision Tree

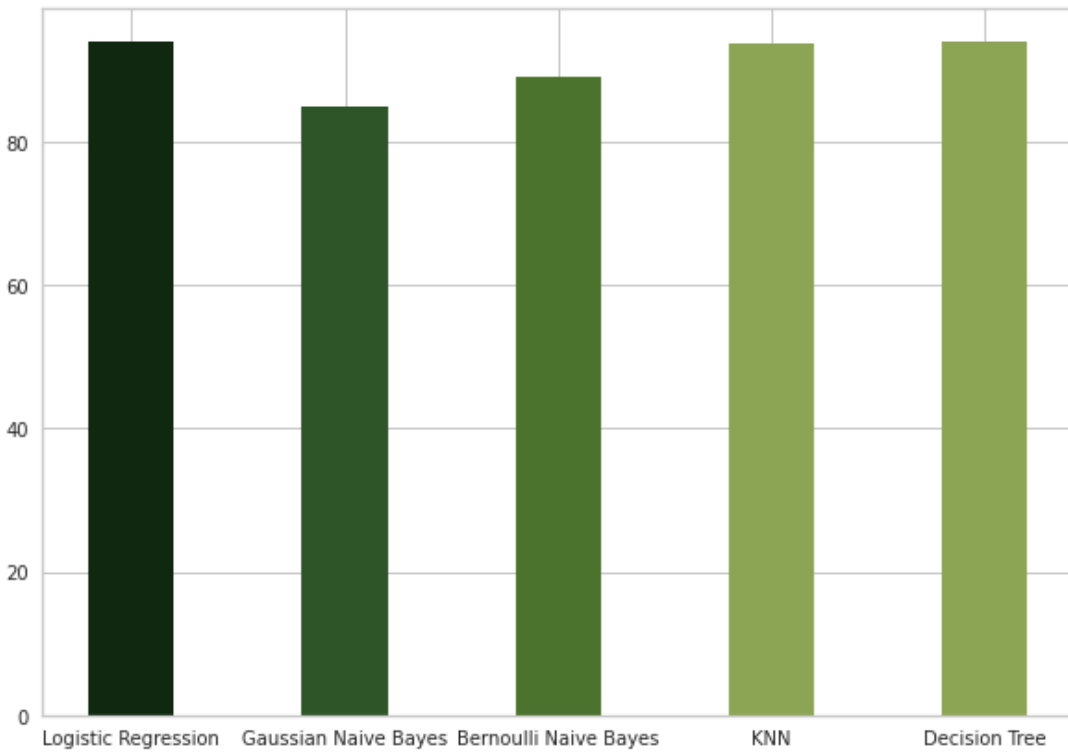
	Precision	Recall	f1-score	support
0.0	0.94	1.00	0.97	19889
1.0	0.00	0.00	0.00	1319
accuracy			0.94	21288
macro avg	0.47	0.50	0.48	21208
weighted avg	0.88	0.94	0.91	21208

In the model summary, we can see that we get a good f1-score - 94% in case of accuracy which is the same as KNN.



In confusion matrix we have 19889 true positives, 0 false negative, 1319 false positive and 0 true negative among 21208 data. We got 93.78% accuracy. Besides, we didn't get any false prediction. If we analyze the ROC curves, we have achieved a good score.

Comparison of machine learning models

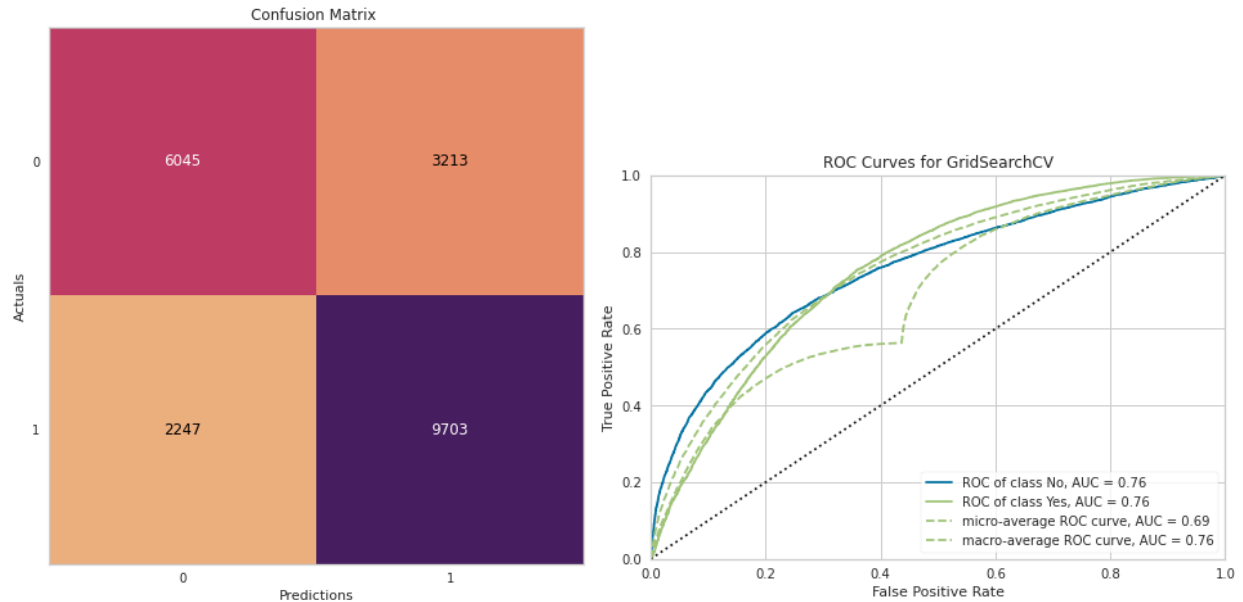


HYPERTENSION:

1. Logistic Regression

	Precision	Recall	f1-score	support
0.0	0.73	0.65	0.69	9258
1.0	0.75	0.81	0.78	11950
accuracy			0.74	21208
macro avg	0.74	0.73	0.73	21208
weighted avg	0.74	0.74	0.74	21208

In the model summary, we got a good f1-score in accuracy. It proves that logistic regression has good accuracy in detecting stroke.

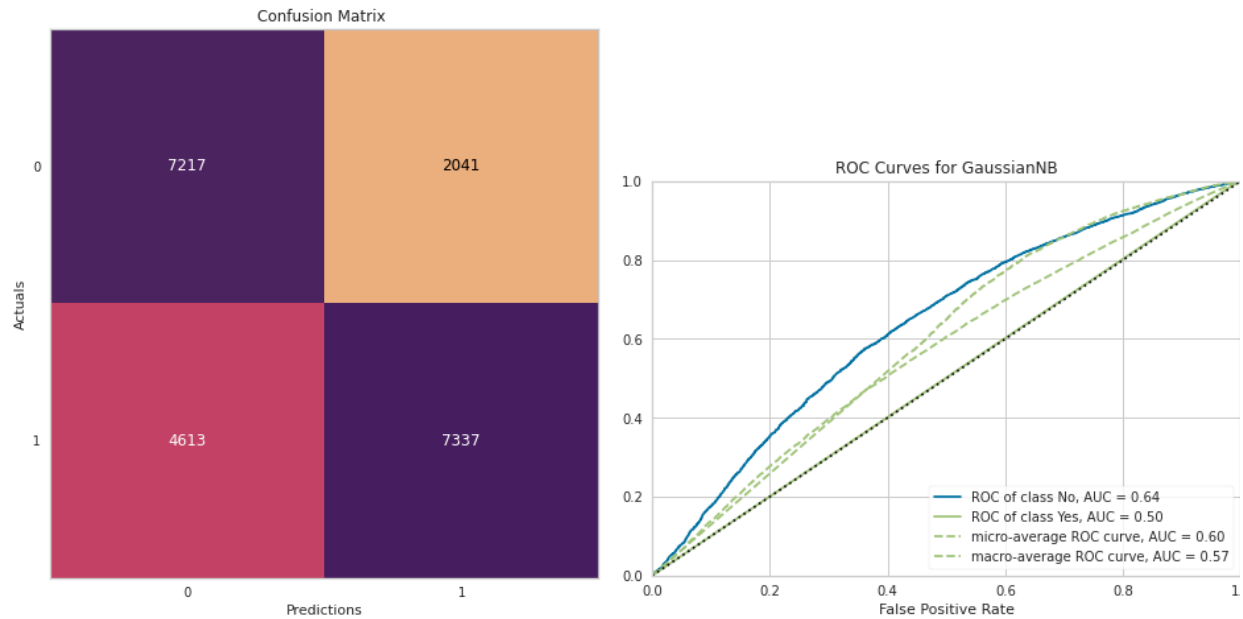


In the confusion matrix we can see among 21208 test data. We got 6045 data as true positive, 3213 data as False negative , 2247 data as false positive and 9703 data as true negative. In ROC curves, we get a good result if we consider the above table.

2. Gaussian Naive Bayes

	Precision	Recall	f1-score	support
0.0	0.61	0.78	0.68	9258
1.0	0.78	0.61	0.69	11950
accuracy			0.69	21208
macro avg	0.70	0.70	0.69	21208
weighted avg	0.71	0.69	0.69	21208

In the model summary, we got a worse f1-score than logistic regression. Since most of the features of our data are dependent, this algorithm didn't work well here.

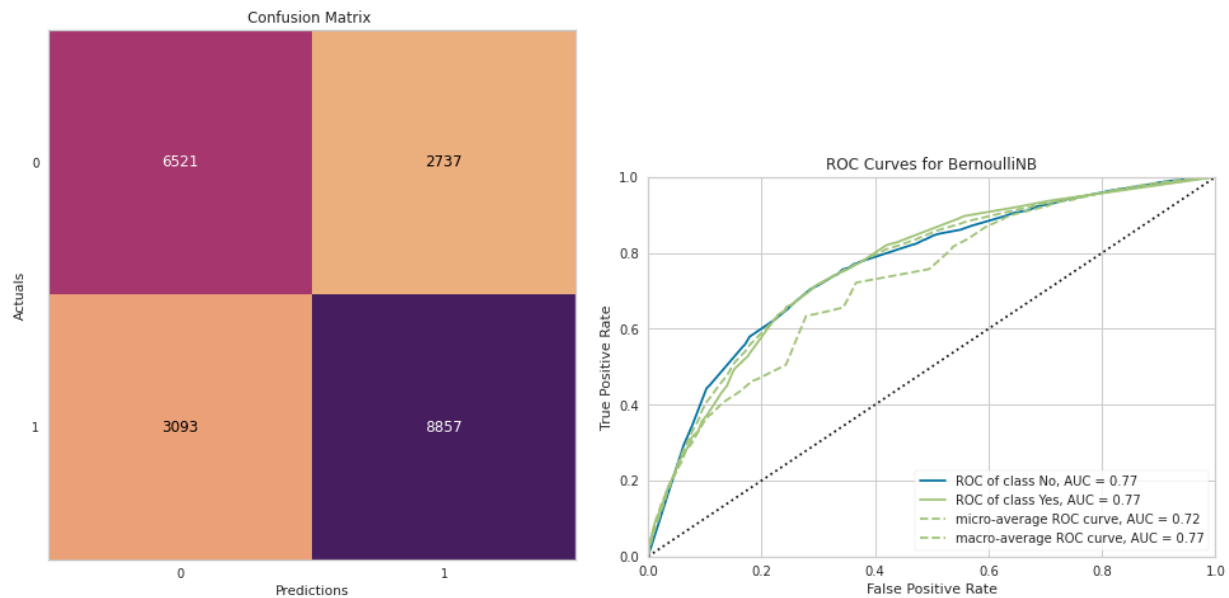


In gaussian naive bayes we got 7217 true positive, 2041 false negative, 4613 false positive and 7337 true negative among 21208 test data. We achieved 0.69% accuracy in detecting hypertension. If we consider ROC curves, we got unsatisfactory results.

3. Bernoulli Naive Bayes

	Precision	Recall	f1-score	support
0.0	0.68	0.70	0.69	9258
1.0	0.76	0.74	0.75	11950
accuracy			0.73	21208
macro avg	0.72	0.72	0.72	21208
weighted avg	0.73	0.73	0.73	21208

In the model summary, we can see that we get a good f1-score - 73% in case of accuracy which is better than gaussian naive bayes.

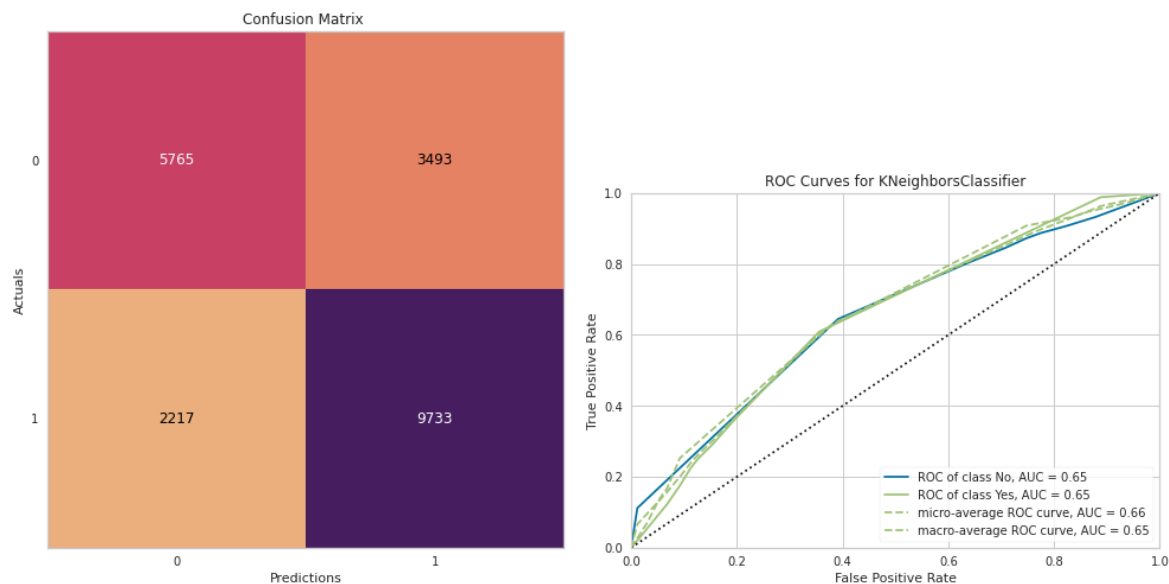


In the confusion matrix we have 6521 true positives, 2737 false negative, 3093 false positive and 8857 true negative among 21208 data. We got 72.51% accuracy. If we analyze the ROC curves, we have achieved a good score.

4) K- Nearest Neighbors

	Precision	Recall	f1-score	support
0.0	0.72	0.62	0.67	9258
1.0	0.74	0.81	0.77	11950
accuracy			0.73	21208
macro avg	0.73	0.72	0.72	21208
weighted avg	0.73	0.73	0.73	21208

In the model summary, we can see that we get a good f1-score - 73% in case of accuracy which is similar to Bernoulli Naive Bayes.

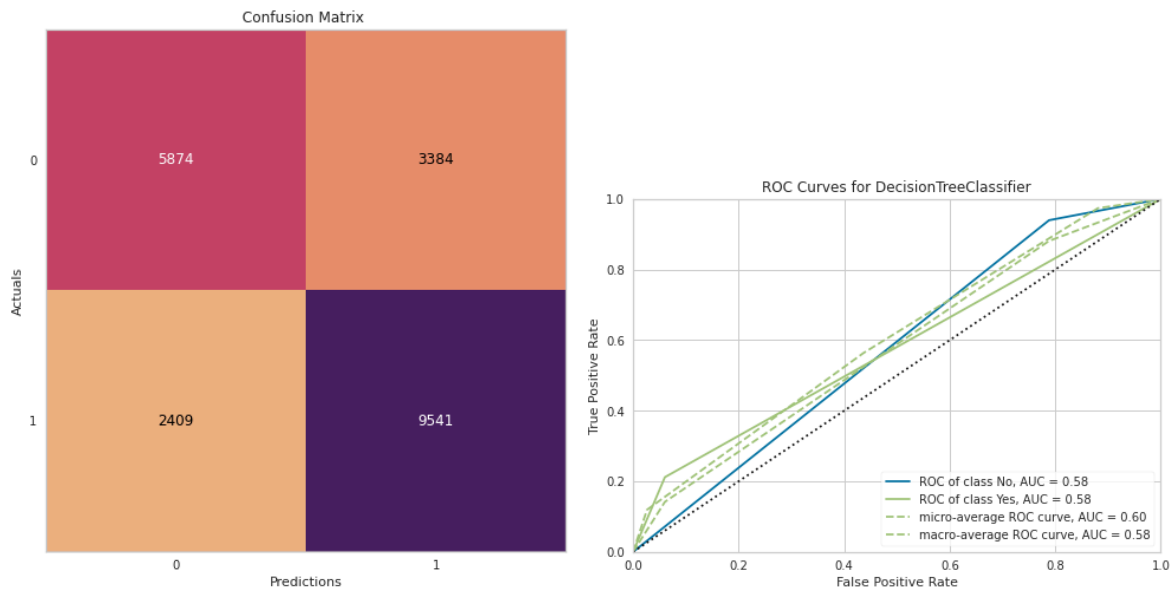


In the confusion matrix we have 5765 true positives, 3493 false negative, 2217 false positive and 9733 true negative among 21208 data. We got 73.08% accuracy. If we analyze the ROC curves, we have achieved an unsatisfactory score.

5) Decision Tree

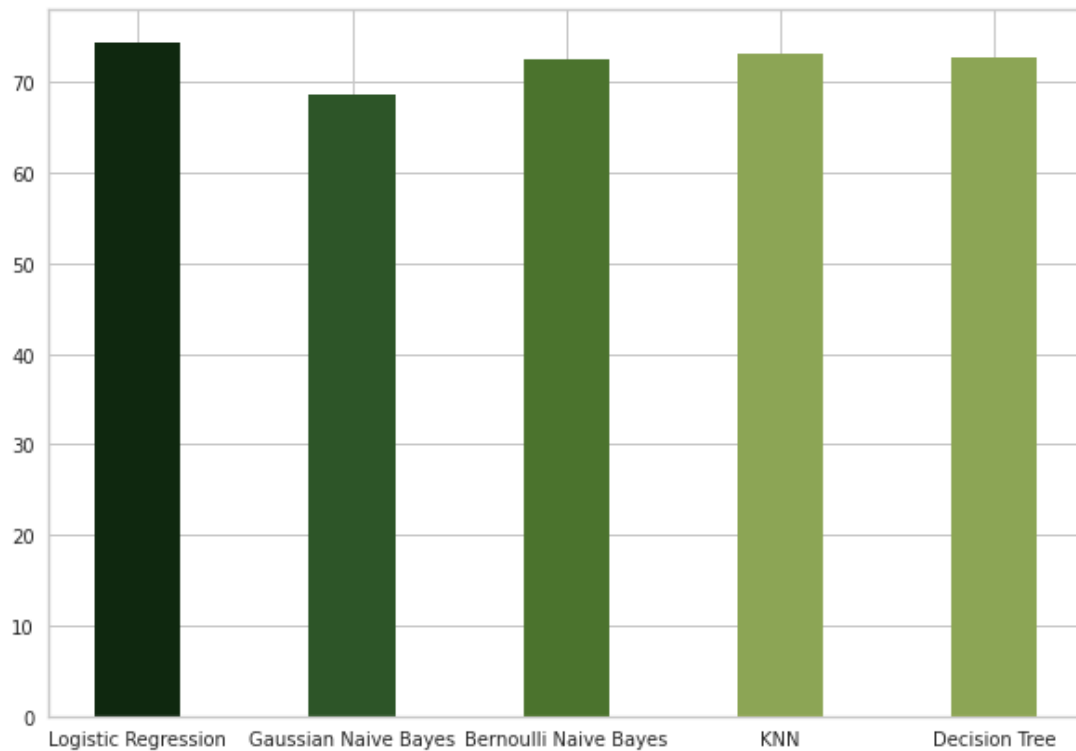
	Precision	Recall	f1-score	support
0.0	0.71	0.63	0.67	19889
1.0	0.74	0.80	0.77	1319
accuracy			0.73	21288
macro avg	0.72	0.72	0.72	21208
weighted avg	0.73	0.73	0.72	21208

In the model summary, we can see that we get a good f1-score - 73% in case of accuracy which is the same as KNN.



In the confusion matrix we have 5874 true positives, 3384 false negative, 2409 false positive and 9541 true negative among 21208 data. We got 72.68% accuracy. If we analyze the ROC curves, we have achieved a good score.

Comparison of machine learning models

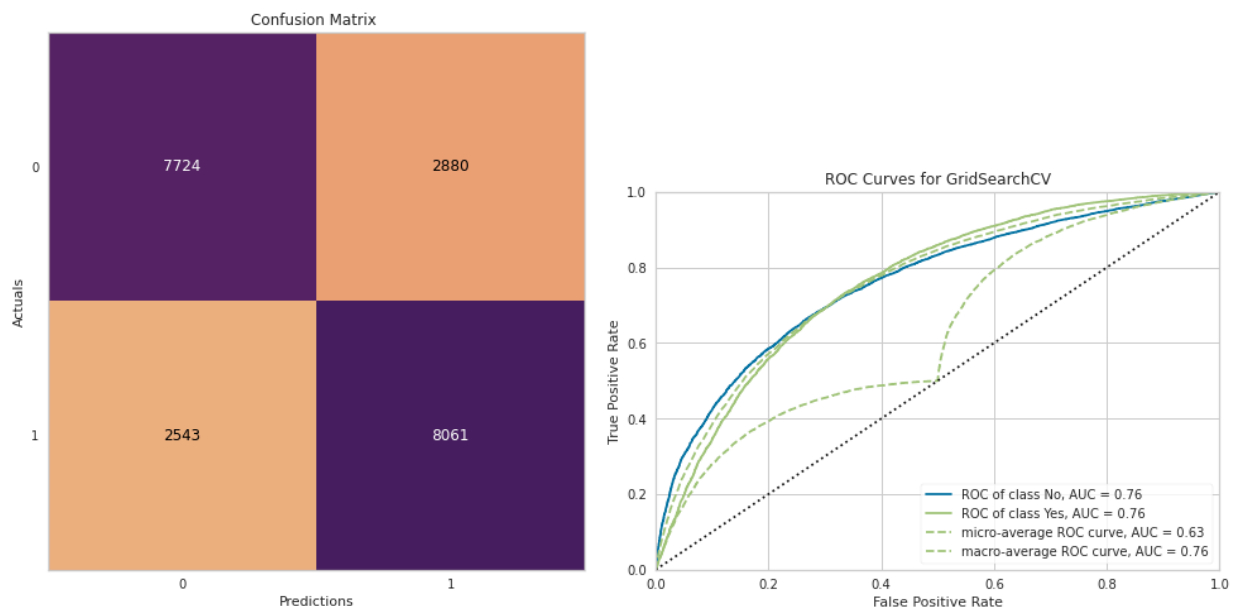


Diabetes:

1) Logistic Regression

	Precision	Recall	f1-score	support
0.0	0.75	0.73	0.74	10604
1.0	0.74	0.76	0.75	10604
accuracy			0.74	21208
macro avg	0.74	0.74	0.74	21208
weighted avg	0.74	0.74	0.74	21208

In the model summary, we got a good f1-score in accuracy. It proves that logistic regression has good accuracy in detecting diabetes.



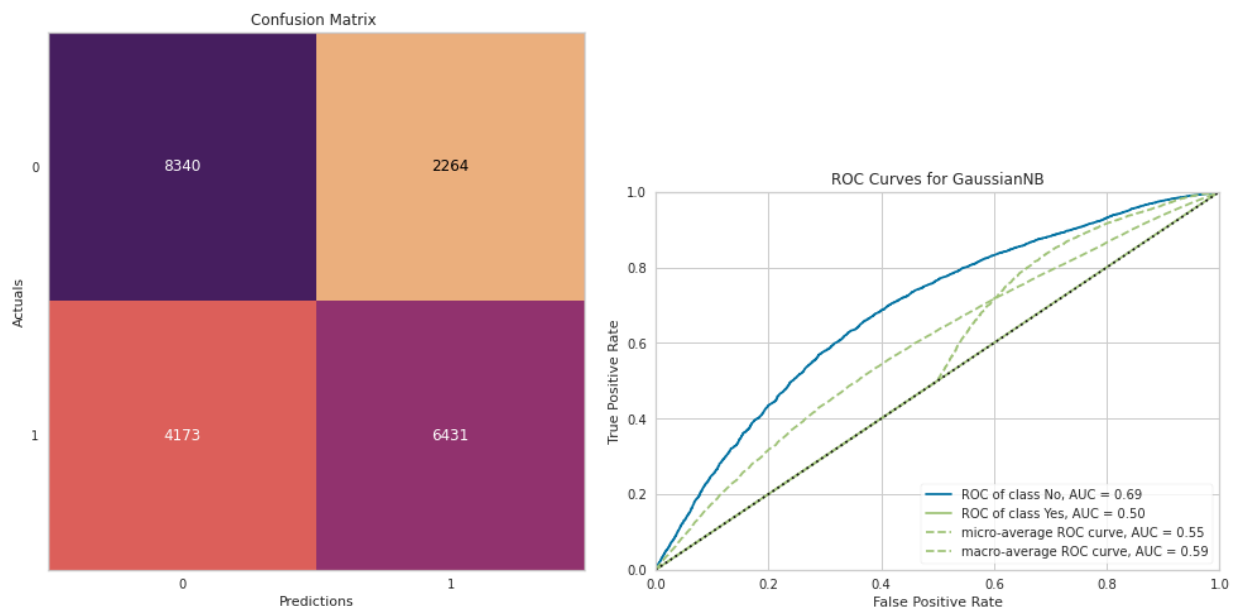
In the confusion matrix we can see among 21208 test data, we got 7724 data as true positive, 2880 data as False negative, 2543 data as false positive and 8061 data as true negative. In ROC curves, we get a good result which is 76% if we consider the above table.

2) Gaussian Naive Bayes

	Precision	Recall	f1-score	support
0.0	0.67	0.79	0.72	10604

1.0	0.74	0.61	0.67	10604
accuracy			0.70	21208
macro avg	0.70	0.70	0.69	21208
weighted avg	0.70	0.70	0.69	21208

In the model summary, we got a good f1-score in accuracy. It proves that Gaussian Naive Bayes has good accuracy in detecting diabetes.

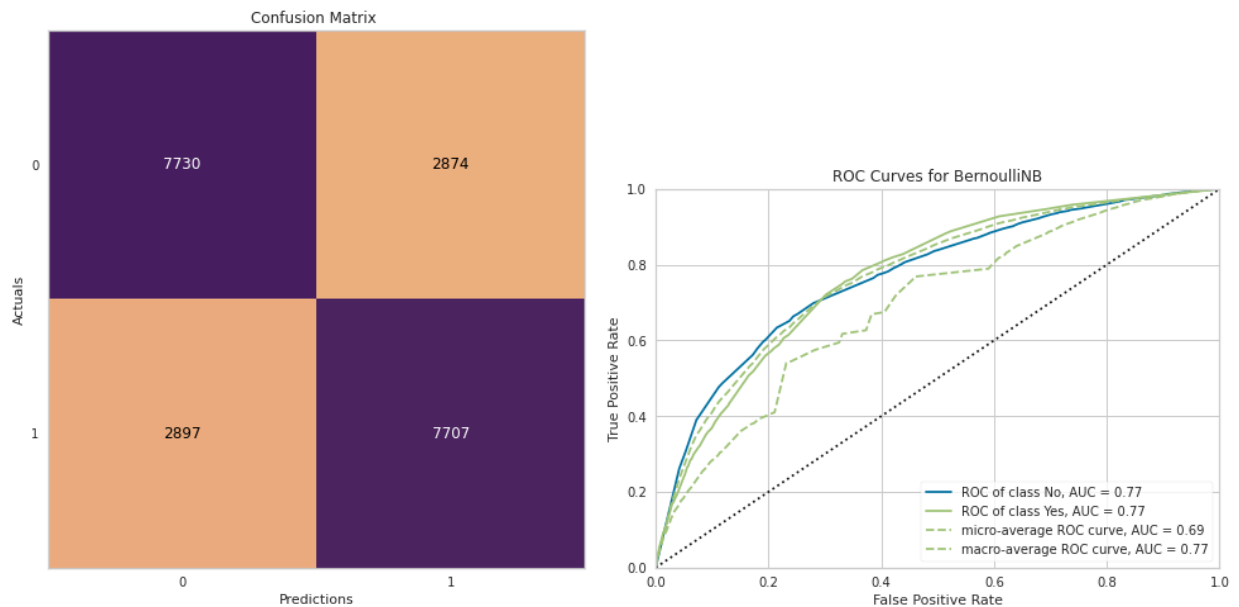


In the confusion matrix we can see among 21208 test data, we got 8340 data as true positive, 2264 data as False negative, 4173 data as false positive and 6431 data as true negative. In ROC curves, we get a satisfactory result which is around 69% if we consider the above table.

3) Bernoulli Naive Bayes

	Precision	Recall	f1-score	support
0.0	0.73	0.73	0.73	10604
1.0	0.73	0.73	0.73	10604
accuracy			0.73	21208
macro avg	0.73	0.73	0.73	21208
weighted avg	0.73	0.73	0.73	21208

In the model summary, we got a good f1-score in accuracy. It proves that Bernoulli Naive Bayes has good accuracy in detecting diabetes.

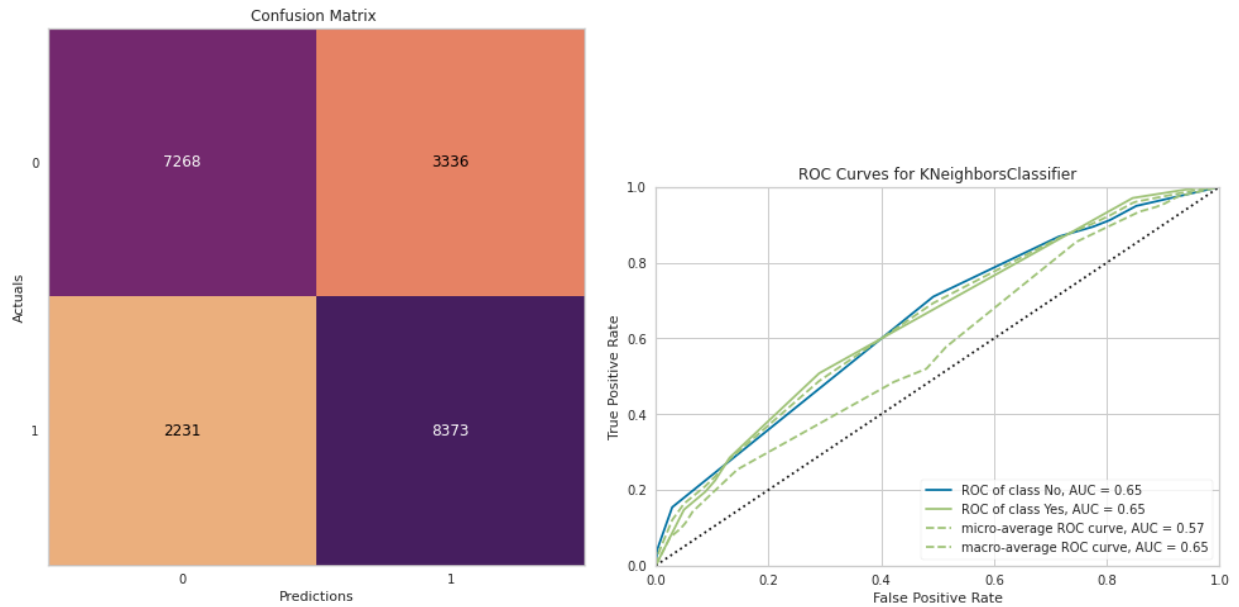


In the confusion matrix we can see among 21208 test data, we got 7730 data as true positive, 2874 data as False negative, 2897 data as false positive and 7707 data as true negative. In ROC curves, we get a good result which is around 77% if we consider the above table.

4) K-Nearest Neighbors

	Precision	Recall	f1-score	support
0.0	0.77	0.69	0.72	10604
1.0	0.72	0.79	0.75	10604
accuracy			0.74	21208
macro avg	0.74	0.74	0.74	21208
weighted avg	0.74	0.74	0.74	21208

In the model summary, we got a good f1-score in accuracy. It proves that K-Nearest Neighbors has good accuracy in detecting diabetes.

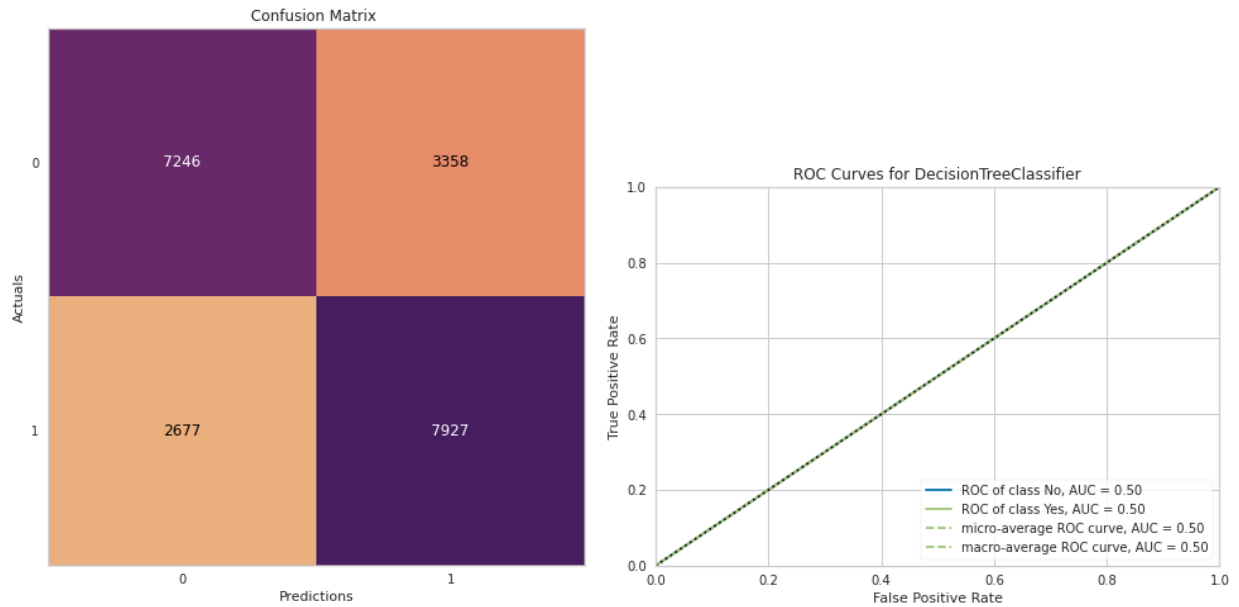


In the confusion matrix we can see among 21208 test data, we got 7268 data as true positive, 3336 data as False negative, 2231 data as false positive and 8373 data as true negative. In ROC curves, we get a satisfactory result which is around 65% if we consider the above table.

5) Decision Tree

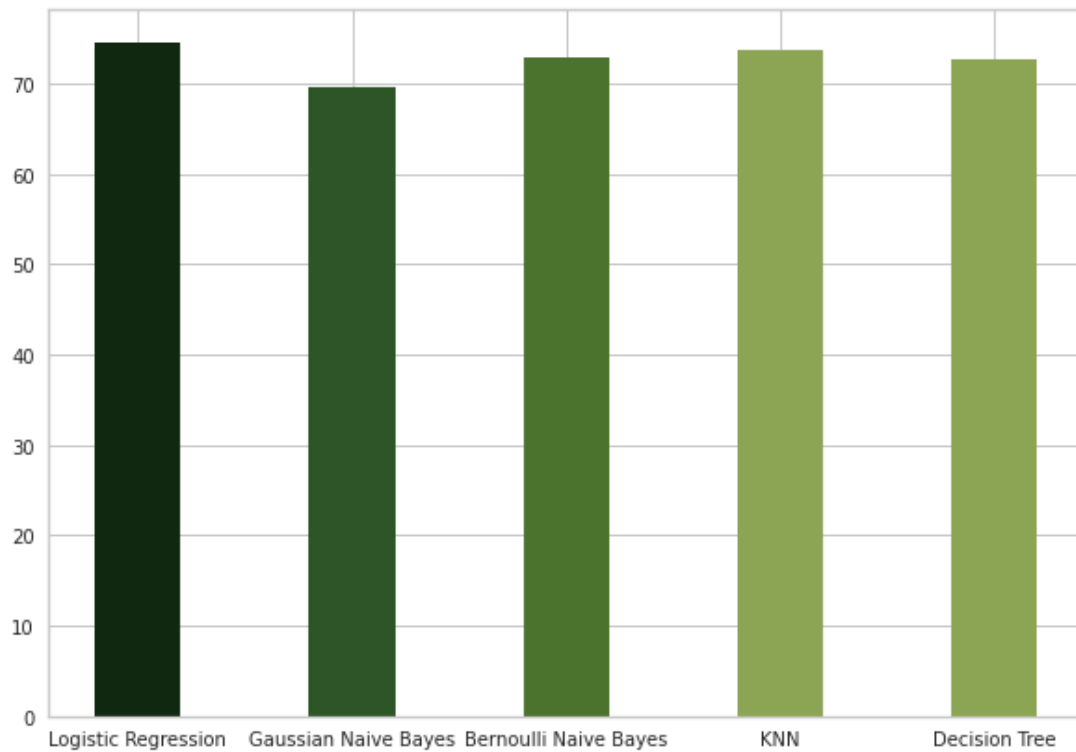
	Precision	Recall	f1-score	support
0.0	0.78	0.64	0.70	10604
1.0	0.69	0.82	0.75	10604
accuracy			0.73	21208
macro avg	0.73	0.73	0.72	21208
weighted avg	0.73	0.73	0.72	21208

In the model summary, we got a good f1-score in accuracy. It proves that Decision Tree has good accuracy in detecting diabetes.

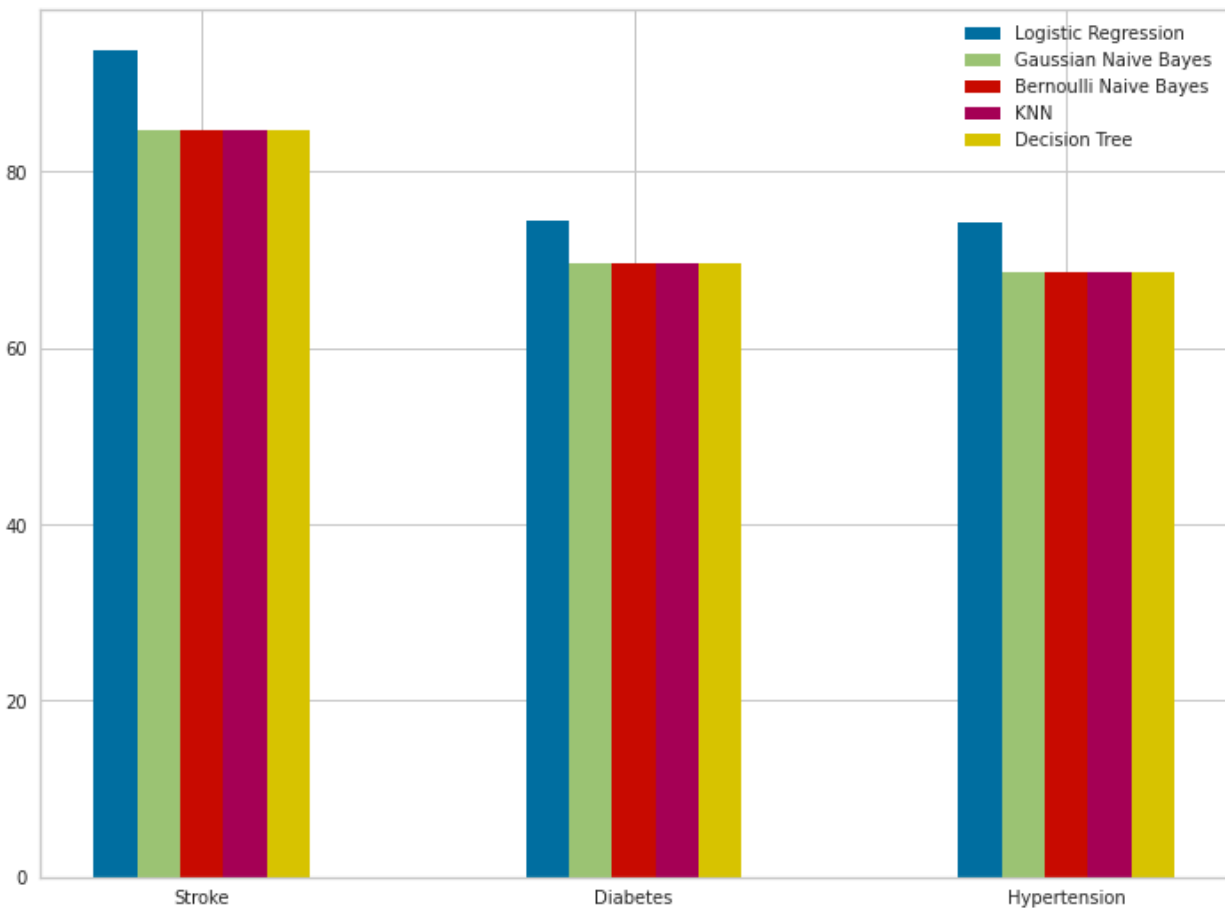


In the confusion matrix we can see among 21208 test data, we got 7246 data as true positive, 3358 data as False negative, 2677 data as false positive and 7927 data as true negative. In ROC curves, we get an average if we consider the above table.

Comparison of machine learning models



Summary of comparison of all the algorithms' accuracy



Conclusion

In conclusion, the use of machine learning techniques in the study of diabetes, strokes, and hypertension has shown promise in improving the diagnosis, treatment, and management of these conditions. ML algorithms have been used to predict the risk of developing these conditions, identify patterns in large datasets, and develop personalized treatment plans. While there are still challenges and limitations to the use of ML in healthcare, it has the potential to significantly improve patient outcomes and overall health outcomes for those with diabetes, strokes, and hypertension. Further research and development of ML techniques in the field of healthcare is likely to continue to yield important insights and advances in the management and treatment of these and other chronic health conditions.